MIND AI × ONTYPOLOGY

Ontypology · Two-Type Application

AI Western Medicine

Consultation & Prescribing

The Complete Method for Accurate Output

Complete Guide to Accurate Output Methods



Space Type × Time Type · Two-Type Driven

Interview Type · Exam Type · Lab Type · Diagnosis Type · Prescription · Follow-up Type

S1-S7 precise prompt pipeline · Four-layer accuracy assurance

End-to-end case: bacterial tonsillitis (antibiotics), full walkthrough

Mind Model reasoning skeleton ¦ × LLM language ability

LONGREAT · the eternal path to AGI

2026

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Before We Begin: Three Conventions

This is a “how to use it” book. It answers one question only: how, with the two Types of Ontypology — the Space Type and the Time Type — as the theoretical basis and Mind AI as the underlying platform, to make every output of AI Western-medicine consultation and prescribing accurate, checkable and traceable.

Before the main topic, three conventions. They run through the whole book and apply to every page.

The Three Conventions of This Book

Convention one · Type. The Chinese 形 is rendered as Type, not Form. Compound terms such as SpecType and DiagnosisType always end in Type.

Convention two · use the theory, do not teach it. This book does not restate the principles of Ontypology (Space Type and Time Type); it only shows how to use them: the three brackets, the intersection criterion, the identity anchor, the three levels of remove, the four-step framework and the Time-Type move tray are all tools taken up and used as they are.

Convention three · cite the platform, do not restate it. This book does not restate the Mind AI platform itself; it cites it only where needed: the Mind Model provides the reasoning skeleton, the LLM provides language ability, and the two carry all output at the base.

Disclaimer (1): This book is a methodological demonstration of AI-assisted consultation and prescribing. All cases, drugs, doses and usages are format demonstrations of output and do not constitute medical advice; for actual diagnosis, treatment and medication, follow the instructions of a qualified physician and local drug regulations.

This book is written output-first: every chapter answers “what to produce, how to produce it, how to accept it”. Readers can move each chapter’s output Types, criteria and prompts straight into their own clinical process and then adjust them to their specialty and regulations.

Table 0-1 Task map: where to find the output you need

What you want to do	Which chapter	The output you will get
Understand how the system is positioned	Chapter 1 (knowing the system)	The medical replica that can be reasoned, where the two Types land, the four questions of accuracy
Know what has to be produced	Chapter 2 (output Types)	Specifications of the six output Types and the prescription BOM
Know how accuracy is assured	Chapter 3 (accuracy assurance)	Four layers of accuracy: criteria, skeleton, review, evidence chain
Walk through one complete encounter	Chapter 4 (six steps)	The six-step method and the three-layer judgement
Get the prompts and examples in hand	Chapter 5 (prompts and case)	The full S1–S7 text + the complete bacterial tonsillitis output
Avoid common errors, assess maturity	Chapter 6 (misuses and checks)	Misuse comparison, ten questions, L1–L4

Introduction: Two Languages, One Stack of Outputs

Before we start, one diagram and two tables lay out the terrain of the whole book: what each of the two languages of Ontypology governs, what they look like when they land on Western-medicine consultation and prescribing, and the three ways to read this book.

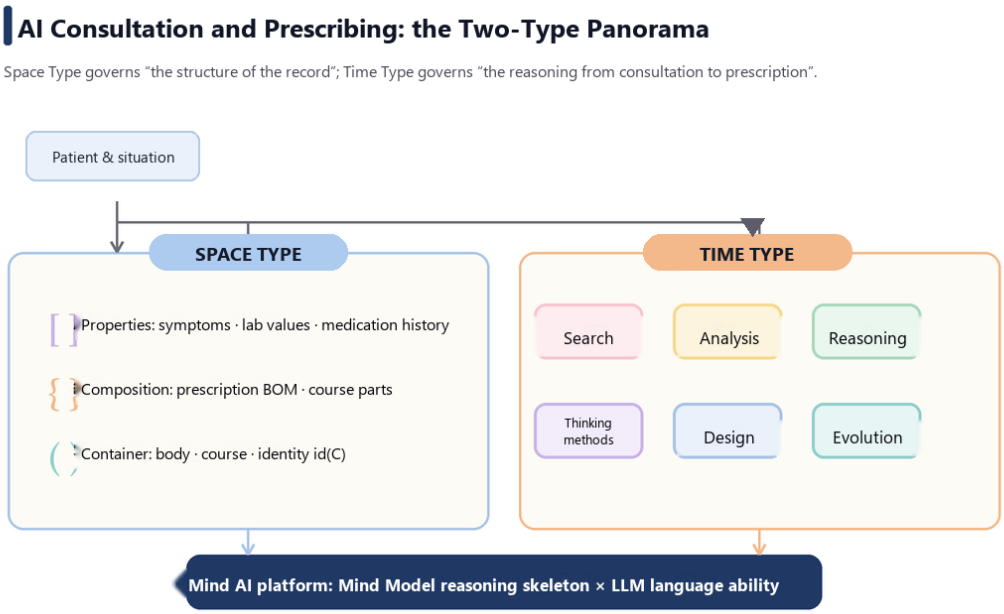


Figure D-1 The Two-Type panorama: the Space Type governs structure, the Time Type governs reasoning, Mind AI carries both at the base

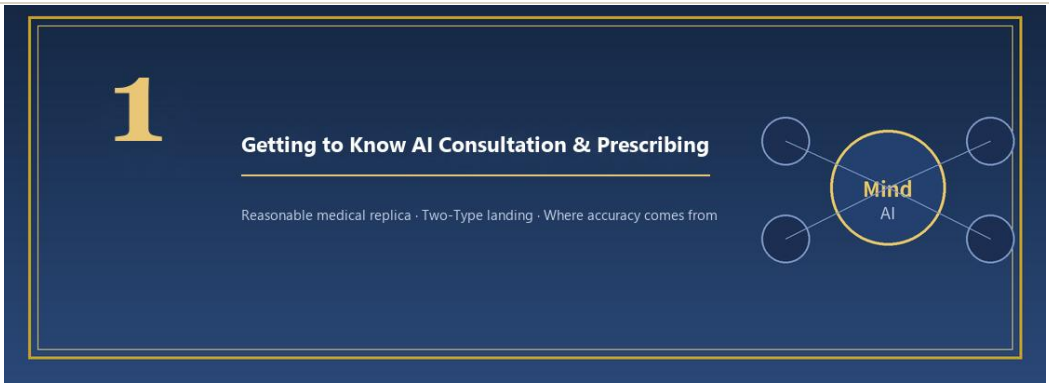
Table D-1 The two languages compared: Ontypology tools × where they land in consultation and prescribing

Ontypology tool	The question it answers	Where it lands in consultation and prescribing
[ ] Property set (abstract)	What kind of thing is this	Fielding of symptoms, lab values and medication history
{ } Composition set (BOM)	What is it made of	Prescription BOM: drug · dose · route · frequency · days
( ) Container set (spatial container)	What is it inside	Body, course time, patient identity id(C)
∩(T) Intersection criterion	What counts as valid	Differential funnel, medication condition criteria
remove, three levels	What happens if a part is removed	Three-level consequences of removing education / the primary drug / the days
The four-step framework	How the situation is identified	Building the situation Type for emergency, outpatient and chronic disease
The Time-Type move tray	Which move this step uses	Search · analysis · reasoning · design · evolution moves
Three-layer judgement	Is · has · right	Three-layer clearance: recognise the disease, assemble the prescription, check the drug

Table D-2 Three ways to read: open the book by your role

Reading path	Who it suits	How to read
The quick read	Decision-makers who want to know what AI can do	Read Chapter 1 (knowing the system) and the maturity section of Chapter 6 (misuses and checks), about 15 minutes
The do-it version	Clinical teams that will actually adopt it	Work through Chapter 4 (six steps) and follow the full S1–S7 text in Chapter 5 (prompts and case) step by step
The audit version	Those responsible for quality and compliance	Accept item by item with the four layers of Chapter 3 (accuracy assurance) plus the ten-question checklist of Chapter 6

Remember one thing before reading on: every output in this book obeys the same iron rule — write the criteria first, then look at the data; have the Type first, then the Number. Reverse the order and every accuracy assurance downstream fails.



Chapter 1 Getting to Know AI Western-Medicine Consultation and Prescribing

Treating “AI consultation and prescribing” as a chatbot Q&A is where most failure stories begin. This chapter repositions it first: AI Western-medicine consultation and prescribing is not a Q&A, it is the maintenance of a medical replica that can be reasoned — and all accuracy grows out of this positioning.

1.1 A Medical Replica That Can Be Reasoned

The value of a record is not that it is “written down” but that it can be “used”. Paper-record thinking writes symptoms as prose, stores diagnoses as impressions and scatters prescriptions across slips; at the return visit everything is asked again, so every encounter starts over in mid-air. A Type-structure replica puts every item where it can be computed: a symptom is a property, a diagnosis has criteria, a prescription has a composition, a return visit is the next page.

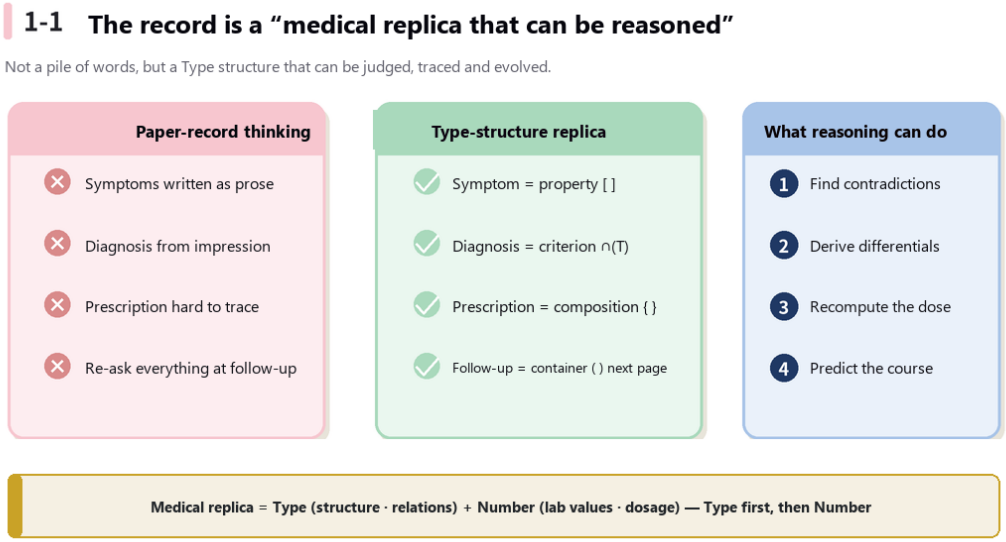


Figure 1-1 From paper record to a replica that can be reasoned: the left can only be read, the right can be reasoned

This replica has to answer four kinds of question: is there a contradiction between data items (for example, self-reported “no allergy” while taking an antihistamine); what can be derived from the available evidence (the differential diagnosis); what happens if one component is removed (a stop-the-drug simulation); and what should be looked at next time (follow-up design). Only when it can answer these four does it deserve the name “reasonable”.

1.2 Type + Number: the Two Materials of a Record

The brain thinks in Type + Number, and a medical record is the same. Number answers “how much”

— temperature 38.4°C, white cells 12,600, a dose of 500mg. Type answers “what it is, what it is made of, where it is, whether it counts” — this is an infection rather than an allergy, this prescription has four components, this course is not finished, this value exceeds the criterion.

Type first, Number second: the number 38.4 only carries clinical meaning once it has been placed inside the Type “fever” (and that Type’s criteria). Every process in this book therefore builds the Type first and fills in the Number afterwards.

Table 1-1 The division of labour between Type and Number in consultation and prescribing

Material	The question it answers	An example in consultation and prescribing
Number	How much	T 38.4°C, HR 92, WBC 12,600, Penicillin V 500mg
Type	What it is · what it is made of	Bacterial tonsillitis, the four components of the prescription BOM, GAS confirmed
Type + Number	The number inside the Type	Centor 3 points (a score of the Type), a 10-day course (a number inside the container)

1.3 How the Two Types Land in Consultation and Prescribing

The Space Type governs the structural side: it splits the record into three axes — property [ ], composition { } and container ( ). The Time Type governs the reasoning side: interview, analysis, reasoning, design and evolution are all verbs, all moves. One complete encounter = one space structure + a string of time moves.

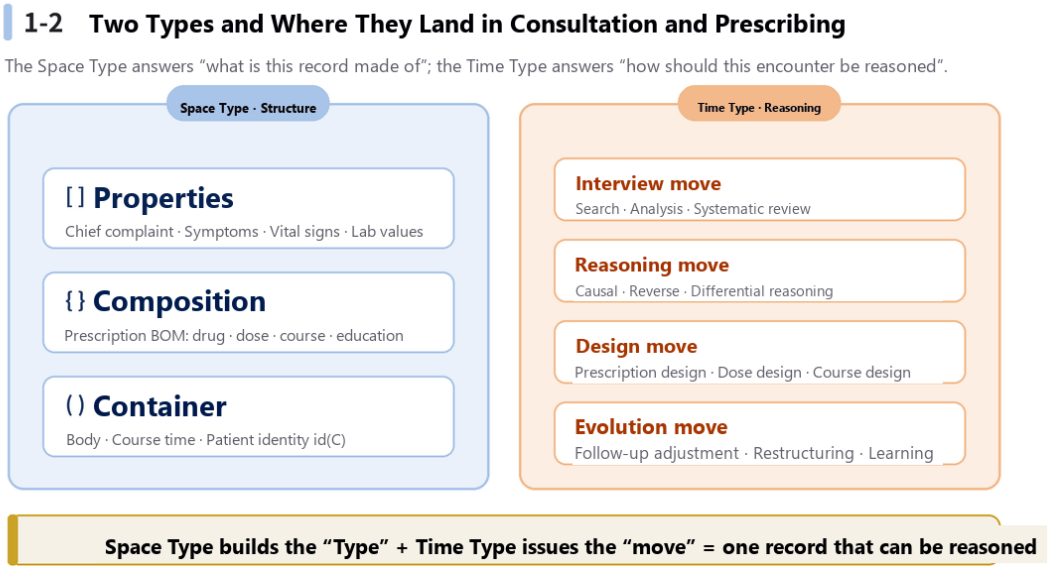


Figure 1-2 Where the two Types land: build the Type on the left, issue the move on the right

Table 1-2 Where the Space Type lands: three brackets × consultation and prescribing

Axis	Set	Where it lands in consultation and prescribing	Criterion
X axis	[ ] property · abstract	Chief complaint, symptoms, vital signs, lab values, medication history	Fields complete, values carry units, abnormalities marked
Y axis	{ } composition · BOM	Prescription BOM: drug, dose, route, frequency, days, education	Every part taken apart knows itself; the primary drug cannot be replaced
Z axis	( ) container	Body, the course time container, patient identity id(C)	Identity stays the same within the course; separate encounters continue as the next page of the same container

Table 1-3 Where the Time Type lands: the four move groups × consultation and prescribing

Move group	Representative moves	Use in consultation and prescribing

Search group	Type-time function search, design search and extraction	Find similar cases in the history base; extract medication rules from guidelines
Analysis group	Analysis, problem solving	Symptom decomposition, lab interpretation, contradiction detection
Reasoning group	Causal reasoning, reverse reasoning, thinking methods	From symptom to cause (causal); from lab result back to the source of infection (reverse)
Design group	Design, function generation, self-evolution	Prescription design, follow-up function generation, therapy restructuring and evolution

1.4 The Mind AI Division of Labour at the Base

This book cites Mind AI but does not restate it: the LLM carries language ability — consultation dialogue, record narrative, patient education text; the Mind Model carries intelligence ability — criterion-based reasoning, BOM structure computation, cross-checking of contraindications, long-term memory and evolution. What consultation and prescribing need is neither “good at talking” nor “good at looking things up”, but the two in one.

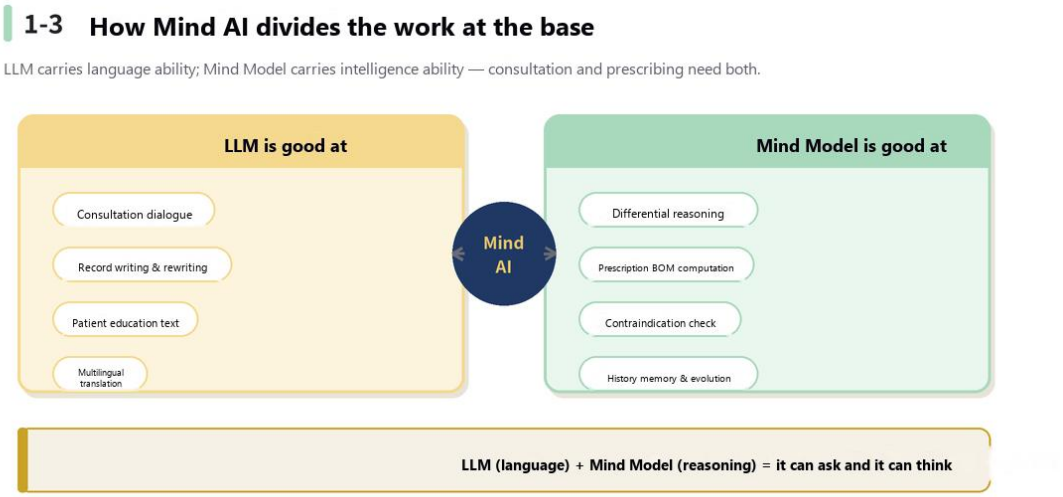


Figure 1-3 The Mind AI division of labour: language belongs to the LLM, intelligence to the Mind Model

1.5 Where “Accuracy” Comes From

Accuracy is not a talent of the model but the discipline of the process. This book splits accuracy into four layers of assurance and sets a gate at each: the first layer, the criterion funnel (∩(T)), decides “what counts”; the second, the deductive skeleton, decides “how the prescription is derived”; the third, the prescription review, decides “whether this drug is right”; the fourth, the Type evidence chain, decides “whether it can be found afterwards”. Only an output that passes all four may leave the door.

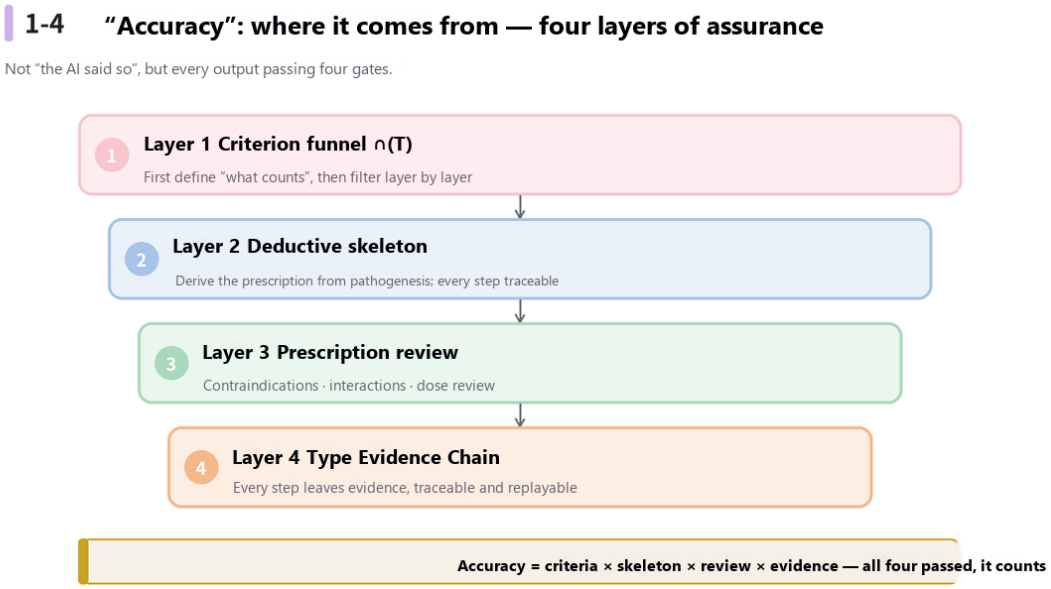


Figure 1-4 Four layers of accuracy: every output has to pass four gates

Table 1-4 The four gates × where they land among the six Types

Gate	What it guards	Which output Types it lands on	Route when it fails
Layer one ∩(T)	What counts as valid	DiagnosisType · LabType	The hypothesis is stopped; go back to selecting the candidate set
Layer two Deductive skeleton	How the prescription is derived	Prescription · DiagnosisType	Walk backwards to find the break, return to S5
Layer three Prescription review	Whether this drug is right	Prescription	Change the drug or adjust the dose, return to S6
Layer four Type evidence chain	Whether it can be found afterwards	All six Types	Complete the chain or void it and re-run

Chapter focus

Chapter focus: (1) AI Western-medicine consultation and prescribing = maintaining a medical replica that can be reasoned; (2) Type first, Number second — every process builds the Type before filling in the Number; (3) the Space Type builds structure, the Time Type issues moves; (4) accuracy comes from four gates, not from the model's confidence.



Chapter 2 The Six Output Types

Every deliverable in the care process has its own Type, its own fields and its own acceptance standard. This chapter sorts the outputs from first visit to follow-up into six Types and shows what they look like on the bacterial tonsillitis case.

2.1 The Six Types at a Glance: One Care Pipeline

The six Types relay in the order of care: InterviewType collects the data, ExamType collects the physical findings, LabType collects the data interpretation, DiagnosisType collects the differential and the confirmation, the prescription collects the medication, FollowType collects what follows. The previous Type is the input of the next; the next is the acceptance of the previous.

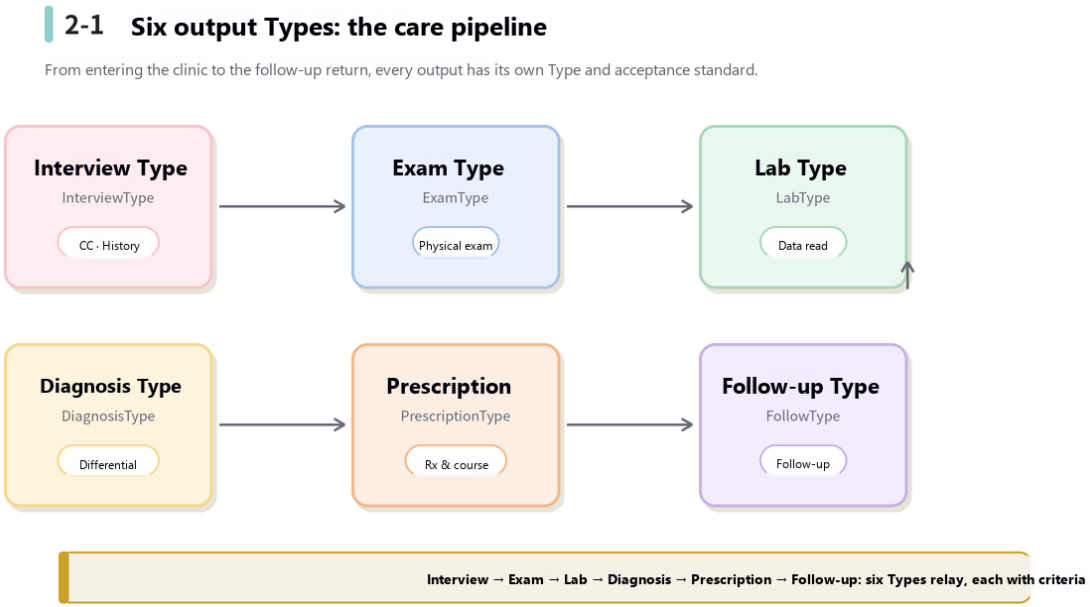


Figure 2-1 The six Types in relay: each Type has its deliverable and its acceptance standard

Table 2-1 Master specification of the six output Types

Output Type	English	Core content	Acceptance standard
Interview Type	InterviewType	Chief complaint, history of present illness, past history, medication history, allergy history, review of systems	Fields complete; the patient's own words retained
Exam Type	ExamType	Vital signs, physical examination findings	Negative findings recorded too
Lab Type	LabType	Test items, values, units, interpretation	Criteria written before interpretation
Diagnosis Type	DiagnosisType	Differential list, criterion scores, confirmed diagnosis	The differential converges from broad to narrow through a funnel

Prescription	PrescriptionType	Prescription BOM, course, education	Every component traceable to the pathogenesis
Follow-up Type	FollowType	Follow-up nodes, assessment items, dose-adjustment conditions	Next page in the same container; identity id(C) unchanged

2.2 InterviewType, ExamType, LabType: Turning "What Was Heard" into "Structure"

The three Types of the first visit turn unstructured "feeling unwell" into computable fields. InterviewType keeps the patient's words and also breaks them into fixed fields; ExamType records the physical findings, especially "what was not found" — negative evidence is extremely valuable in a differential diagnosis; LabType is "write the criteria first, then look at the data", and adding criteria after seeing the data is forbidden.

2-2 Interview Type · Exam Type · Lab Type: turning "what was heard" into "structure"

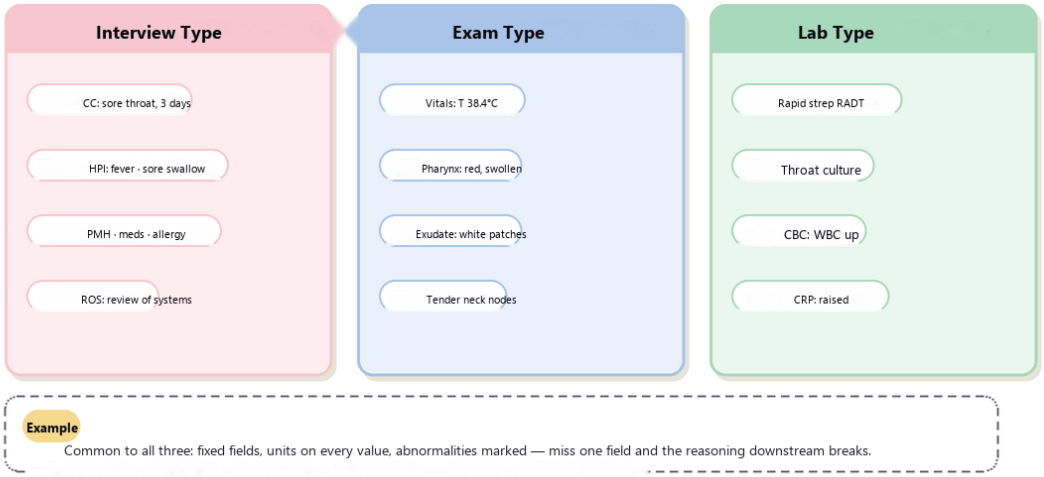


Figure 2-2 The first three Types: the fields and requirements of interview, exam and lab

Table 2-2 InterviewType field card (InterviewType)

Field	Content example (Ms Chen, 32)	Requirement at the Type level
Chief complaint CC	Sore throat for 3 days	The patient's own words + duration
History of present illness HPI	Worse on swallowing, fever to 38.4°C, no cough	Aggravating / relieving factors mandatory
Past history PMH	No chronic disease; no recurrent tonsillitis	Negatives recorded too
Medication history	Occasional paracetamol; no long-term prescription drugs	Includes OTC and supplements
Allergy history	Reports no drug allergy	Explicitly record "yes / no / unknown"
Review of systems ROS	No rash, no joint pain, no asthma	Systems relevant to the differential must be asked

2.3 DiagnosisType: Differential Is a Funnel, Not Guesswork

The core of DiagnosisType is a funnel: the candidate set has to be broad enough, and convergence has to rest on criteria. Take a sore throat: the first layer separates infection from non-infection, the second splits infection into viral and bacterial, the third scores with the Centor criteria, and the fourth confirms with RADT (rapid antigen detection test). Every layer has to record "on what grounds it was filtered out".

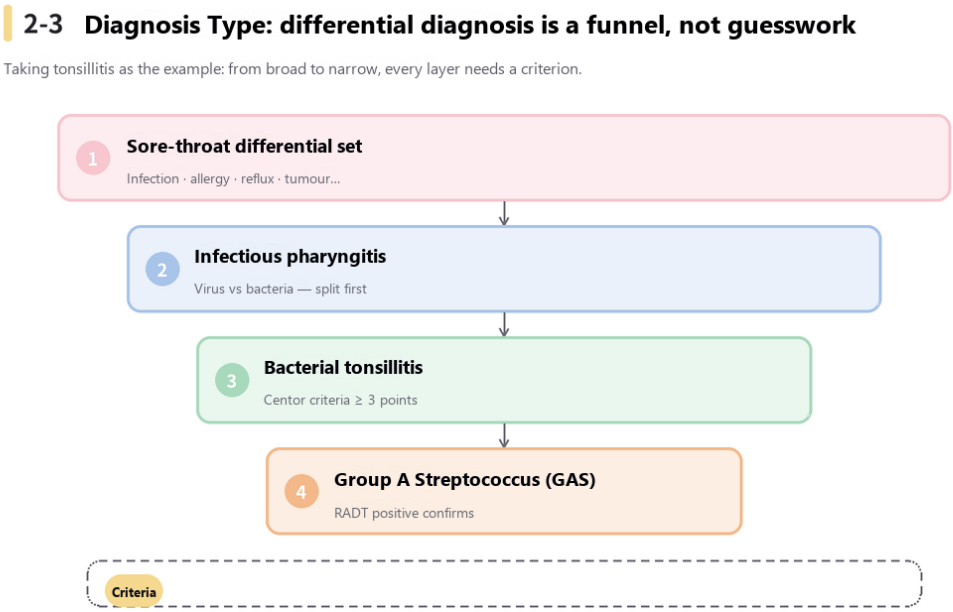


Figure 2-3 The diagnostic funnel: every layer of convergence has a criterion you can check

Table 2-3 Differential comparison table (sore throat · excerpt)

Candidate diagnosis	Supporting evidence	Opposing evidence	Criterion result
Viral pharyngitis	Sore throat, fever	No cough, RADT positive	Excluded
Bacterial tonsillitis (GAS)	Centor 3, exudate, RADT positive	None	Confirmed
Infectious mononucleosis	Young, sore throat	No lymphadenopathy, no splenomegaly	Excluded · keep under observation
Peritonsillar abscess	Severe unilateral pain	No trismus, no muffled voice	Excluded

2.4 Prescription: One Prescription Slip Is One BOM

A prescription is a composition set { }: taken apart, every piece knows itself. In this case the prescription BOM has four components — the primary drug, symptom relief, local treatment and education — and each has its own role and its own non-substitutability.

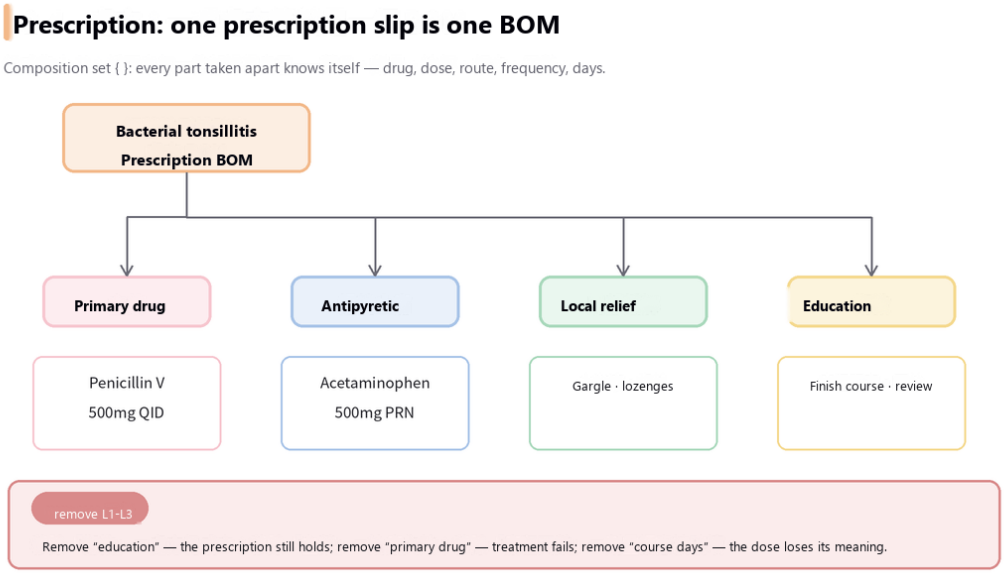


Figure 2-4 The prescription BOM: four components and the three-level consequences of remove

Table 2-4 Prescription BOM specification (bacterial tonsillitis · demonstration only)

Component	Drug / content	Dose · route · frequency	Days	Role

Primary drug	Penicillin V	500mg · PO · QID	10 days	Bactericidal · not to be replaced at will
Symptom relief	Acetaminophen	500mg · PO · PRN (≤ QID)	As needed	Antipyretic and analgesic · supportive
Local treatment	Warm saline gargle	Every 4–6 hours	10 days	Relief · supportive
Education	Finish the course · when to return	—	10 days	Guarantees the course is completed

remove three levels × prescription

The three levels of remove applied to a prescription: remove the “education” component and the prescription still holds (level one: quality affected); remove the “primary drug” and the treatment fails (level two: function destroyed); remove the “course days” field and the meaning of the dose collapses, so the prescription cannot be executed (level three: meaning destroyed). When writing a prescription, level-three components must be marked in bold and never omitted.

2.5 FollowType: the Course Is a Container, a Return Visit Is the Next Page

FollowType hangs on the container ( ): Day 0 to Day 30 are the same course-time container, and the patient identity id(C) does not change inside it — a new name, a new number or a referral does not change the identity. A return visit is not a fresh start but a new time slice of the same chain: only the differences are appended, never the whole text recopied.

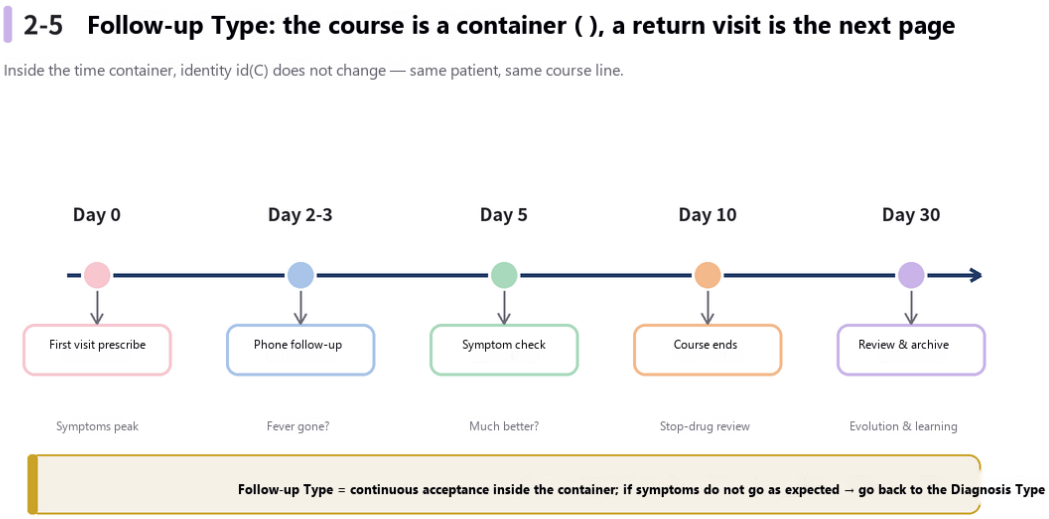


Figure 2-5 The follow-up timeline: five nodes inside the container

Table 2-5 Follow-up node specification (FollowType)

Node	Assessment items	Pass standard	Route when it fails
Day 2–3 phone follow-up	Whether the fever has gone, change in pain	Fever gone · pain decreasing	Go back to DiagnosisType and re-judge
Day 5 symptom assessment	Swallowing, energy, appetite	Clearly improving	Bring the visit forward · consider complications
Day 10 course ends	Whether the course was completed	All medication taken · symptoms relieved	Re-educate those who did not finish
Day 30 review	Any recurrence	No recurrence	Repeated episodes → refer to ENT

Chapter focus

Chapter focus: (1) the six Types relay, the previous is input and the next is acceptance; (2) negative evidence must be recorded as well as positive; (3) diagnosis is a criterion funnel, not intuition; (4) a prescription is a BOM, and level-three components cannot be omitted; (5) follow-up continues as the next page inside the container, identity id(C) unchanged.



Chapter 3 Four-Layer Accuracy Assurance

"Can you trust what the AI prescribes" is the life-or-death question for this system. The answer does not depend on how big the model is but on how strict the gates are. This chapter splits accuracy into four layers and explains, layer by layer, how each works on the tonsillitis case.

3.1 Layer One · Criterion Funnel ∩(T)

The spirit of the intersection criterion ∩(T) is: data ∩ criteria, and a hypothesis that fails the criteria may not go further. The practical order must never be reversed — write the criteria down first, then let the data in; if you look at the data first and then fit the criteria to it, that is hindsight, not reasoning.

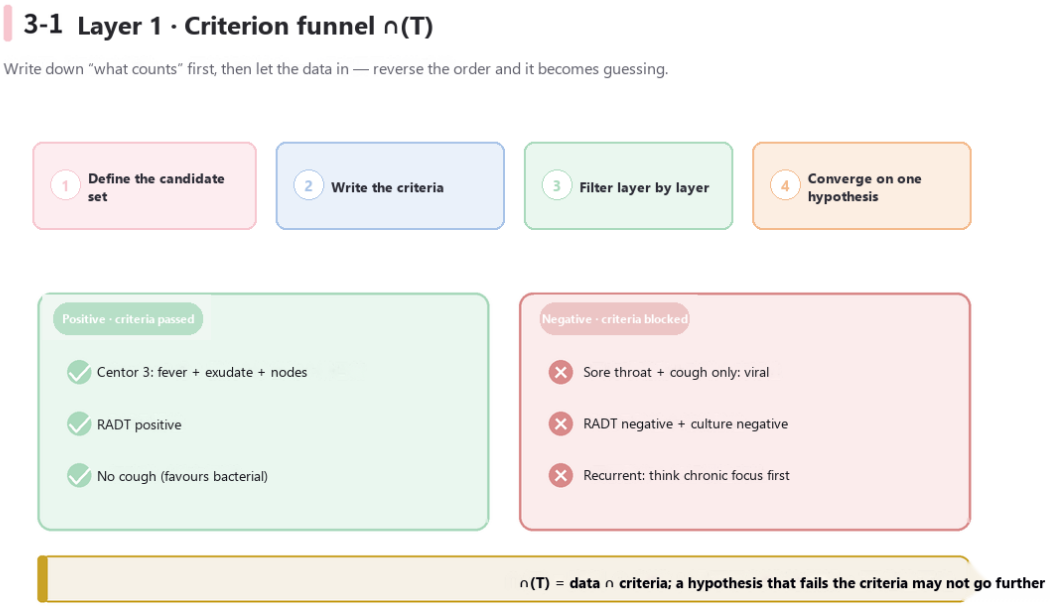


Figure 3-1 The criterion funnel: establish the criteria first, then filter the data

Table 3-1 The ∩(T) criterion table · tonsillitis differential

Criterion	How it is measured	Pass threshold	What it blocks
Centor criteria	One point each for fever, absence of cough, cervical nodes, exudate	≥ 3 points enters the bacterial hypothesis	Viral pharyngitis
RADT rapid test	Throat swab antigen test	Positive confirms GAS	Non-streptococcal infection
Course criterion	Days of symptoms and trend	> 3 days and worsening	Mild self-limiting sore throat
Red-flag criterion	Dyspnoea, trismus, drooling	Any one present → emergency referral	Abscess · epiglottitis

3.2 Layer Two · Deductive Skeleton

A prescription has to grow along a visible chain of reasoning: cause → pathogenesis → evidence → principle → prescription. This chain is the deductive skeleton — it turns “prescribing” from table lookup into derivation. The check is simple: walk it backwards, and if any link does not match, the output goes back.

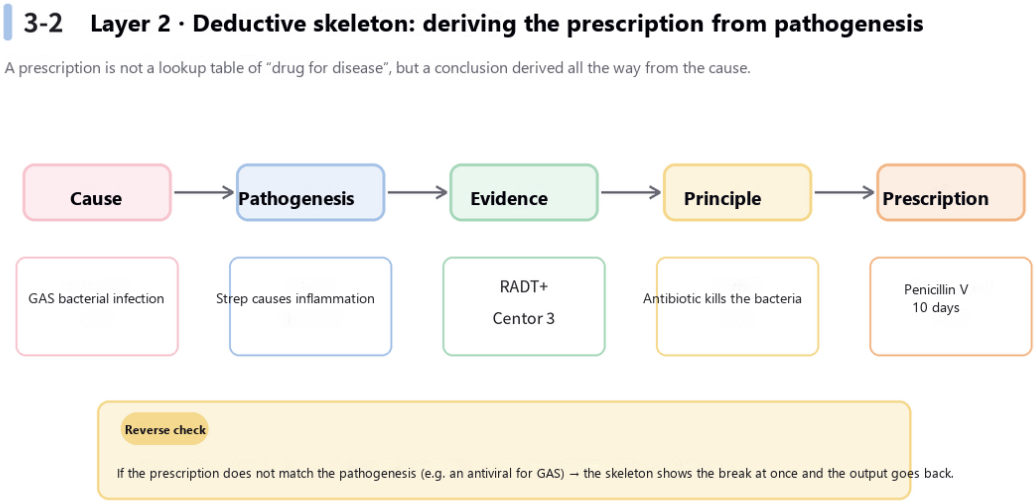


Figure 3-2 The deductive skeleton: from cause to prescription, link by link

Table 3-2 The deductive skeleton · tonsillitis case

Link	Content	Corresponding output Type
Cause	Group A Streptococcus (GAS) infection	DiagnosisType
Pathogenesis	The bacterium causes acute suppurative inflammation of the pharynx	DiagnosisType
Evidence	Centor 3 + RADT positive + raised WBC	ExamType · LabType
Principle	Antibiotic bactericidal + symptom relief	DiagnosisType
Prescription	Penicillin V 500mg QID × 10d	Prescription

3.3 Layer Three · Prescription Review

Even when the diagnosis is right and the drug is right, the prescription has to pass one last gate before it leaves: the three questions of the prescription review — contraindications, interactions, dose. All three must answer “yes” for release; any “no” means changing the drug or going back to re-derive.

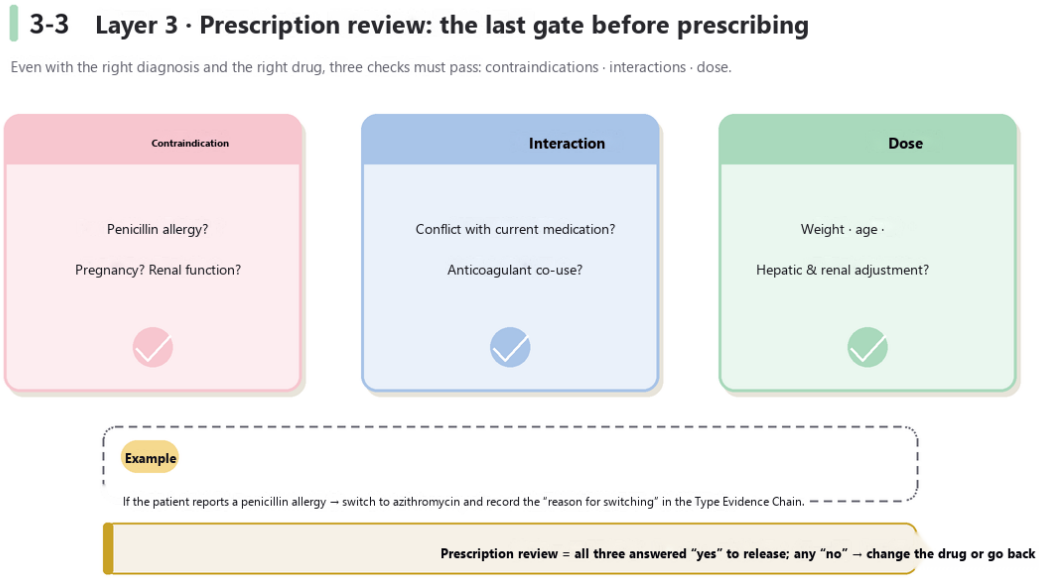


Figure 3-3 The three questions of the prescription review: the last gate before prescribing

Table 3-3 Prescription review checklist · this case

Check item	What is checked	Result in this case	Route
Contraindication	Penicillin allergy history, pregnancy and breastfeeding, renal function	Allergy history negative · not pregnant or breastfeeding · renal function normal	Passed
Interaction	Conflict with current medication (e.g. anticoagulants)	No long-term medication	Passed
Dose	Adjustment for weight, age, hepatic and renal function	32 years, 56kg, standard adult dose	Passed
Fallback plan	Alternative drug in case of allergy	Azithromycin, 5-day course (fallback)	Reserve

Disclaimer (2): Table 3-3 and every drug dose in this book are output demonstrations only and do not constitute medical advice; an actual prescription must be decided by a qualified physician according to the individual patient, local drug regulations and the latest guidelines.

3.4 Layer Four · Type Evidence Chain

The Type evidence chain requires: every input has a source, every process has grounds, every output can be audited, and any moment can be replayed. In space, each output carries its evidence; in time, each change leaves a record. The two are cross-accepted and together form complete traceability.

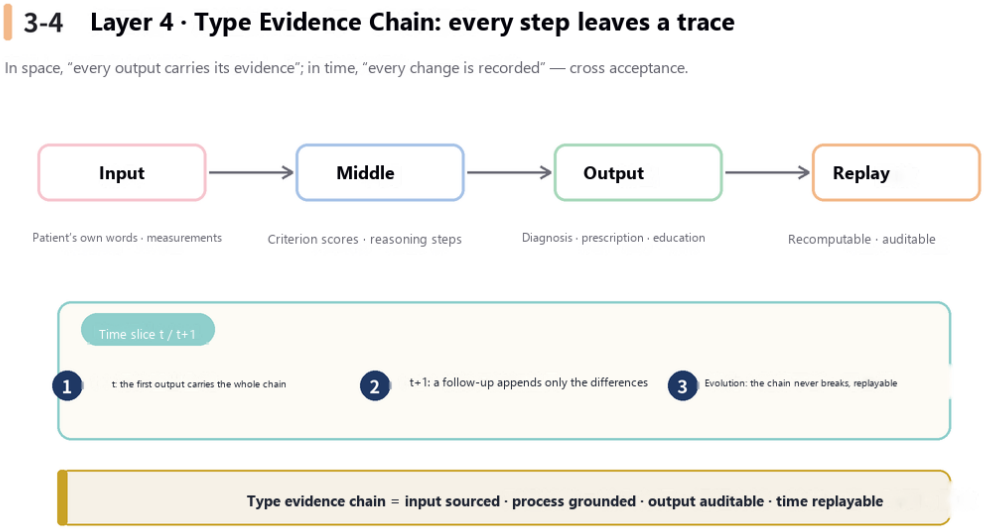


Figure 3-4 The Type evidence chain: four stages of trace plus time slices

Table 3-4 The Type evidence chain · field specification

Chain stage	What is kept	Where it is stored	What it is used for in audit
Input	The patient's own words, measurements, lab reports	InterviewType · ExamType · LabType	Replay what was relied on at the time
Middle	Criterion scores, reasoning steps, excluded hypotheses	DiagnosisType	Recompute whether the reasoning holds
Output	Diagnosis, prescription, education, review report	Prescription	Trace the source of the prescription
Replay	Differences at the return visit, reasons for dose change, efficacy	FollowType	Evolution and learning of the therapy

3.5 Space × Time Cross Acceptance

The four gates act in both directions at once: in space they accept "is this output structurally complete", in time they accept "does this change have a reason". If either direction breaks, the whole chain fails.

Table 3-5 Cross acceptance table

Direction	Acceptance question	Example of failure
Space · structure	Are the six Types' fields complete? Can the BOM parts be taken apart? Does the evidence attach?	The prescription has no days field; a lab value has no unit
Space · meaning	Are the level-three components there? Is the naming consistent?	The primary drug is written as "drug A" and cannot be traced
Time · order	Criteria before data? Skeleton before prescription?	Prescribe first and add the diagnostic reasons afterwards
Time · evolution	Is there a reason for the dose change? Does the container continue?	The return visit starts from scratch and the identity chain breaks

Chapter focus

Chapter focus: (1) ∩(T) establishes the criteria before filtering the data; (2) the prescription must be walkable backwards along the deductive skeleton to the cause; (3) all three questions of the prescription review must pass for release; (4) the Type evidence chain guarantees that any moment can be replayed; (5) space × time cross acceptance — missing either fails.



Chapter 4 Using It Accurately · the Six Steps

The first three chapters gave the positioning, the output Types and the gates; this chapter strings them into one operable route: the six steps. Each step has an input, an output and an acceptance, and the chapter shows how to select, assemble and swap moves in two situations — acute infection and chronic disease.

4.1 The Four-Step Framework: Identify the Situation First

The four-step framework is the preparation before building the Type: identify the situation first, then fix the boundary, then set the fields, and finally set the criteria. If the situation is wrong, everything after it is wrong — the emergency situation and the chronic dose-adjustment situation use completely different moves.

Table 4-1 The four-step framework × consultation and prescribing

Step	Action	Demonstration on the tonsillitis case
Identify the situation	Emergency / outpatient / chronic? Acute or chronic?	Outpatient · acute infection · single episode
Fix the boundary	The scope and exclusions of this encounter	Limited to GAS pharyngitis; abscess and systemic disease excluded
Set the fields	The field list of the six Types in place	Six interview fields · four exam items · three lab items
Set the criteria	∩(T) and the red-flag criteria first	Centor ≥ 3, RADT, red flag → immediate referral

4.2 The Six Steps End to End

The six steps in order are: identify the situation, build the Types by axis, select the expansion pattern, generate the output, accept against the criteria, archive and evolve. The first three are preparation, the fourth is production, the fifth is the gate, and the sixth makes the system more accurate over time.

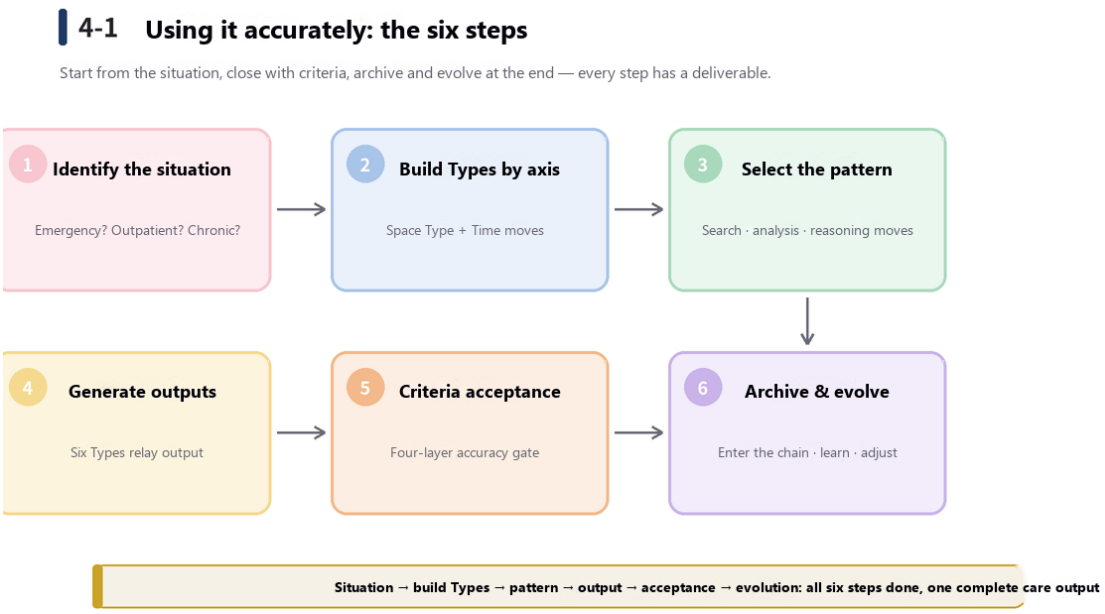


Figure 4-1 The six steps: every step has a deliverable

Table 4-2 Six-step operation card

Step	Input	Action	Output	Acceptance
1 Identify the situation	Chief complaint	The four-step framework	Situation tag	The situation matches the chief complaint
2 Build Types by axis	Situation tag	Build the Types with the three brackets	Six-Type skeleton	Fields complete
3 Select the expansion pattern	Six-Type skeleton	Choose moves from the move tray	Move list	The moves match the situation
4 Generate the output	Move list	The S1–S7 pipeline	Six-Type output	Format compliant
5 Accept against the criteria	Six-Type output	The four layers of accuracy	Accepted output	All four layers passed
6 Archive and evolve	Accepted output	Enter the chain · learn	Time slice	The chain is continuous

4.3 Three-Layer Judgement: Is · Has · Right

The three-layer judgement of Type logic, applied to consultation and prescribing, becomes three questions: what is it (same-or-different judgement), what does it have (composition judgement), is it right (criterion judgement). The three pass from left to right, and only then does the output hold.

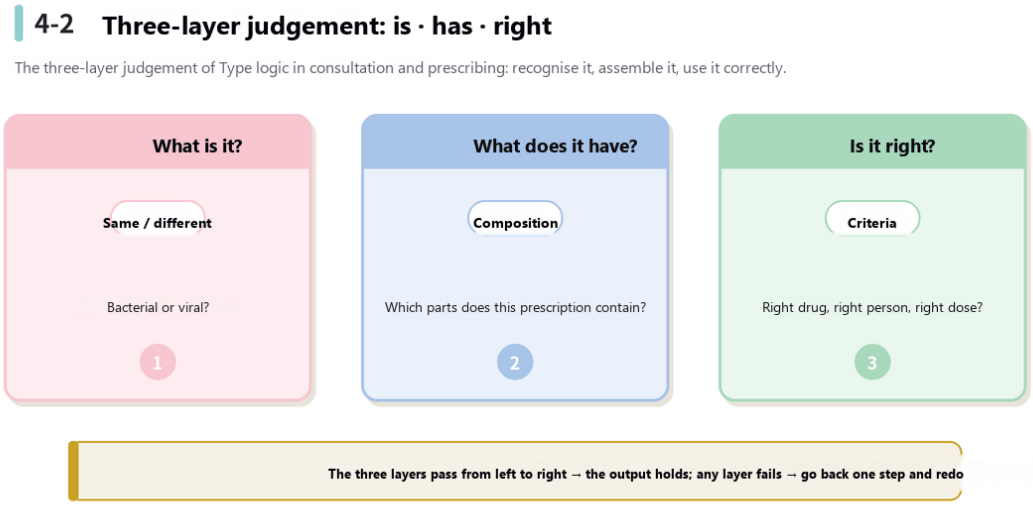


Figure 4-2 The three-layer judgement: recognise it, assemble it, use it correctly

Table 4-3 The three-layer judgement · worked on the case

Layer	Question	Answer in this case	If it fails
Is	Bacterial or viral?	Bacterial (Centor 3 + RADT+)	Back to S4 to re-run the differential funnel
Has	Which components does the prescription contain?	Primary drug, relief, local, education — four	Back to S5 to complete the BOM
Right	Right drug for the person, the disease, the dose?	Penicillin V passes the three questions	Back to S6 to change the drug or adjust the dose

4.4 Selecting · Assembling · Swapping Moves

The Time Type is a board of verbs: in the same patient situation, acute infection uses fast moves (search + causal reasoning + prescription design) and chronic dose adjustment uses long moves (history replay + statistical reasoning + ladder adjustment). Moves can be combined and can be swapped midway — when follow-up goes worse than expected, stop the “design move” first and go back to the “reasoning move” to re-differentiate.

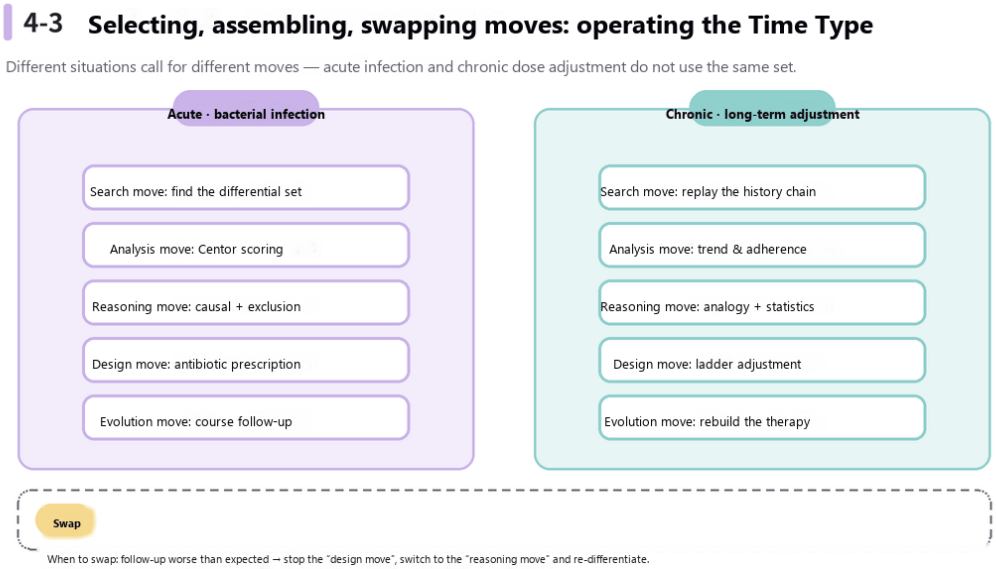


Figure 4-3 Move arrangement in the two situations

Table 4-4 Selecting, assembling and swapping moves compared

Action	Acute infection situation	Chronic dose-adjustment situation

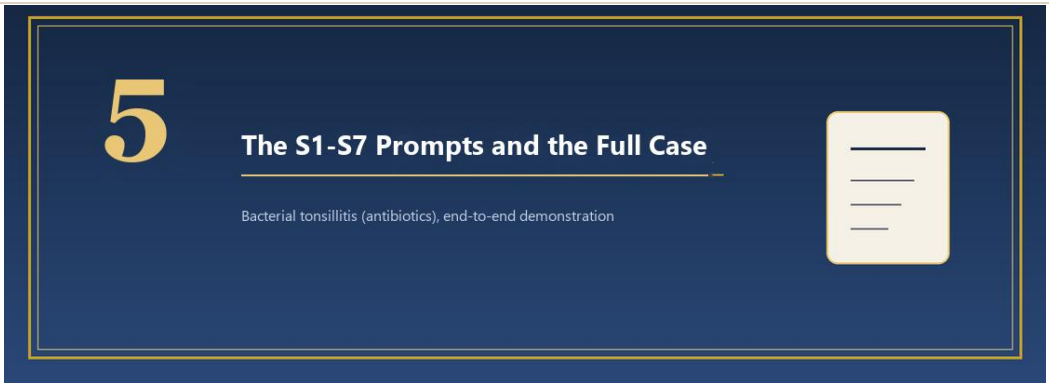
Selecting moves	Search move (find the differential set) + reasoning move (causal)	Search move (replay the history chain) + reasoning move (statistical)
Assembling moves	Analysis + reasoning + design: score → confirm → prescribe	Analysis + design + evolution: trend → ladder → adjust
Swapping moves	No fever resolution at 48–72h → swap to a reasoning move and re-differentiate	Target not met in three months → swap to a design move and change to second-line drugs

Table 4-5 The six steps × misuse warning points

Step	The most common shortcut	Warning signal
1 Identify the situation	Starting without separating acute from chronic	The complaint says “recurrent” or “for years” yet an acute move is used
2 Build Types by axis	Leaving fields blank to fill in later	More than two “to be filled” cells in the output table
3 Select the expansion pattern	Using only the moves you are familiar with	A chronic case with no replay move at all
4 Generate the output	Jumping straight to S5 for the prescription	The prescription has no corresponding DiagnosisType input
5 Accept against the criteria	The review report says only “passed”	The three questions have no reason field of their own
6 Archive and evolve	Closing the case without entering the chain	The next first visit starts asking from zero again

Chapter focus

Chapter focus: (1) the four-step framework comes first — a wrong situation makes everything wrong; (2) every one of the six steps has an input, an output and an acceptance; (3) the three-layer judgement “is · has · right” passes layer by layer; (4) moves can be selected, assembled and swapped, and a swap needs a trigger condition.



Chapter 5 The S1–S7 Prompts and the Full Case

This chapter delivers two directly usable things: the full text of the seven prompts, and a complete worked output from consultation to follow-up. The case continues the bacterial tonsillitis example that runs through the book — Ms Chen, 32, sore throat for three days with fever.

5.1 Pipeline Design: A Seven-Stage Relay

Each of the seven prompts has its own job: S1 history taking, S2 examination planning, S3 lab interpretation, S4 differential diagnosis, S5 prescription generation, S6 prescription review, S7 education and follow-up. The output of one stage is the input of the next, and any stage can be audited or re-run on its own.

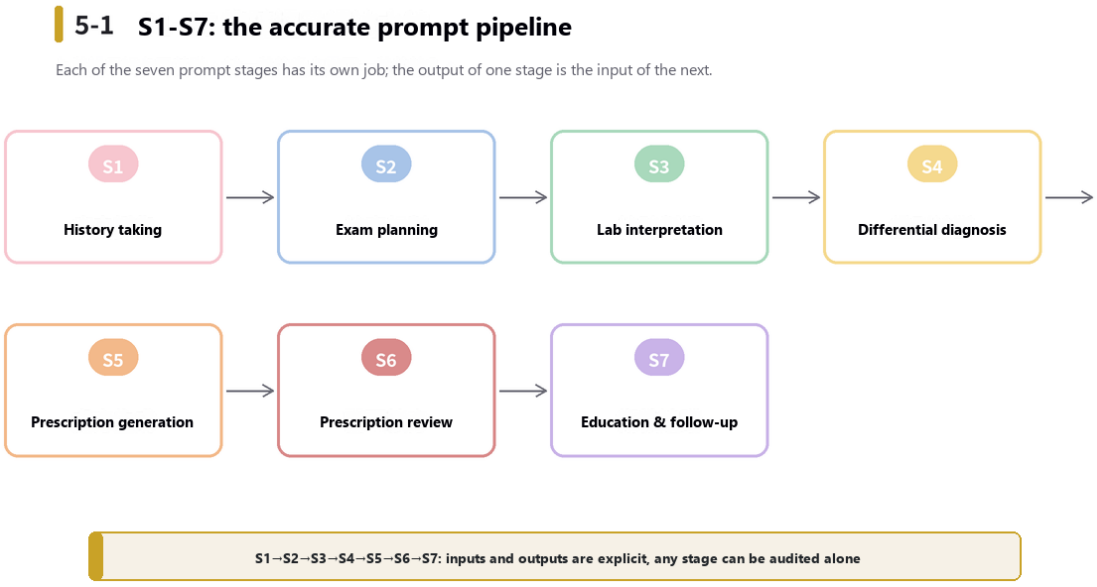


Figure 5-1 The seven-stage relay: inputs and outputs clear, auditable stage by stage

Table 5-1 S1–S7 role table

Stage	Role	Input	Output Type
S1	History taking	Patient statement	InterviewType
S2	Examination planning	InterviewType	ExamType (planning)
S3	Lab interpretation	Lab data + criteria	LabType
S4	Differential diagnosis	InterviewType · ExamType · LabType	DiagnosisType
S5	Prescription generation	DiagnosisType	Prescription
S6	Prescription review	Prescription	Review report
S7	Education and follow-up	Prescription + review report	FollowType

5.2 The Full Text of the Seven Prompts

Below are prompts that can be copied and used directly. Items in 〔 〕 are variables to replace at use; every prompt has built-in format constraints and an honesty clause: “if unsure, say unsure”.

S1 History Taking

You are a Western-medicine outpatient physician. Conduct a structured consultation with the patient and produce the “InterviewType”.

Fields: chief complaint CC (patient's words + days), history of present illness HPI (aggravating / relieving factors), past medical history PMH, medication history Med (including OTC), allergy history Allergy, review of systems ROS.

Requirements: (1) ask only 1–3 questions at a time, never all at once; (2) record negative evidence too; (3) fields the patient does not provide are filled “not provided”, and guessing values is forbidden.

Output format: a Markdown table with columns “field | content | data source (patient's words / physician's judgement)”.

S2 Examination Planning

Input: InterviewType 〔paste the S1 output〕.

Task: plan this physical examination and produce the “ExamType” plan sheet.

Requirements: (1) list the mandatory items and “what is to be confirmed by examining it”; (2) list the items the differential needs but that are not examined this time, with reasons; (3) red-flag items (dyspnoea, difficulty opening the mouth, neck mass) must be listed.

Output format: table “examination item | purpose | differential hypothesis served | red-flag link”.

S3 Lab Interpretation

Input: ExamType 〔paste the S2 output〕 + lab data 〔paste the report〕.

Task: write the criteria first, then look at the data, and produce the “LabType” interpretation sheet.

Requirements: (1) for every lab item write the “criterion (threshold and direction)” before the value and the conclusion; (2) values must carry units; (3) items beyond the criterion are marked abnormal, stating which hypothesis they support or oppose.

Output format: table “item | criterion | value | conclusion | hypothesis supported / opposed”.

S4 Differential Diagnosis

Input: InterviewType + ExamType + LabType 〔paste the S1–S3 output〕.

Task: produce the “DiagnosisType”.

Requirements: (1) the differential list runs from broad to narrow with at least 3 candidates; (2) every candidate carries supporting and opposing evidence; (3) converge layer by layer through the criterion funnel, stating at each layer on what grounds the rest were filtered out; (4) the confirmed diagnosis carries a confidence level; (5) negative red flags are written too.

Output format: first the funnel table “layer | criterion | candidates | result”, then the conclusion “confirmed diagnosis | confidence | still to be confirmed”.

S5 Prescription Generation

Input: DiagnosisType 〔paste the S4 output〕 + patient baseline data (age, weight, hepatic and renal function, allergy history, current medication).

Task: produce the “PrescriptionType” BOM.

Requirements: (1) BOM components: primary drug, symptom relief, local treatment, patient education; (2) each item states dose, route, frequency, days and total quantity; (3) the primary drug is marked “not freely replaceable” and the days are marked “level-three component”; (4) the primary drug carries a deductive-skeleton cross-reference (cause → principle → drug); (5) an alternative plan is reserved (in case of allergy).

Output format: prescription slip table + skeleton cross-reference table; end with a disclaimer.

S6 Prescription Review

Input: prescription 〔paste the S5 output〕 + patient baseline data.

Task: run the three prescription-review questions and produce the review report.

The three questions: (1) contraindication check (allergy history, pregnancy and lactation, hepatic and renal function, disease contraindications); (2) interaction check (current medication, food, laboratory interference); (3) dose check (adjustment for weight, age, hepatic and renal function).

Requirements: each question gives one of three states — pass / caution / block — with a reason; any “block” must come with an alternative drug and an alternative dose; release is forbidden unless all three pass.

Output format: table “check item | what is checked | result | reason | pathway”.

S7 Education and Follow-Up

Input: prescription + review report 〔paste the S5–S6 output〕.

Task: produce the “FollowType”.

Requirements: (1) every follow-up node (Day 2–3 / Day 5 / Day 10 / Day 30) carries assessment items and a pass standard; (2) every node states its “failing pathway” (which step to push back to and re-judge); (3) the education text comes in two versions, patient (plain language) and physician (brief); (4) the return red-flag list must be attached.

Output format: node table + two education versions + red-flag list.

5.3 Full Case Output

Below are excerpts of the actual results produced for Ms Chen's case along S1–S7. The output trace of the whole pipeline is shown in the figure, then the details stage by stage:

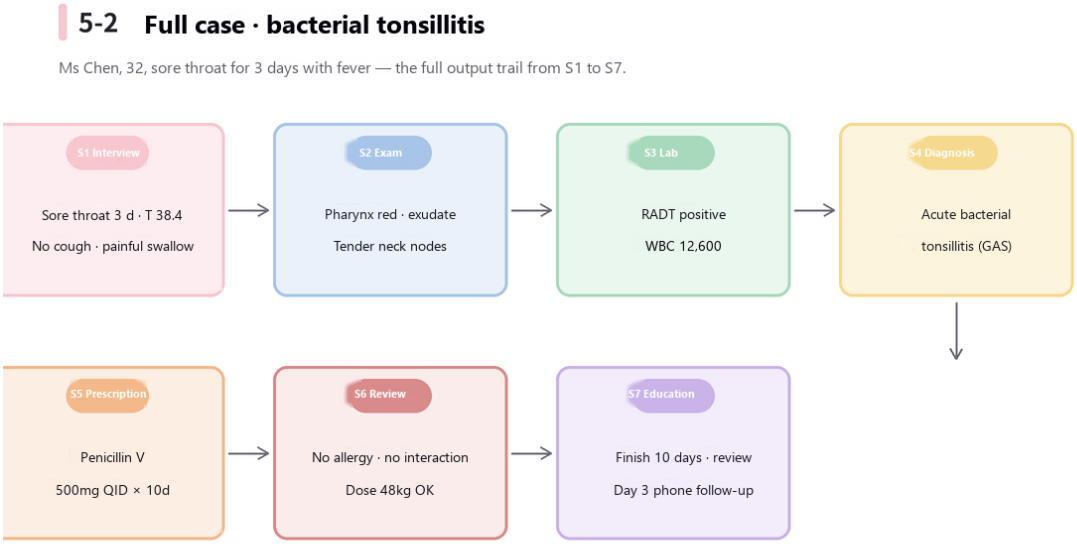


Figure 5-2 Case output trace: the seven-stage relay from consultation to education

First look at the InterviewType and the two Types of examination and laboratory:

Table 5-2 Case output excerpt (1): InterviewType

Field	Content	Source
Chief complaint CC	Sore throat for 3 days	Patient's own words
History of present illness HPI	Worse on swallowing, fever to 38.4°C, no cough	Patient's own words
Past medical history PMH	No chronic disease, no recurrent tonsillitis	Patient's own words
Medication Med	Occasionally takes paracetamol 500 mg	Patient's own words
Allergy Allergy	Reports no drug allergy	Patient's own words
ROS	No rash · no arthralgia · no wheeze	Consultation confirmation

Table 5-3 Case output excerpt (2): ExamType · LabType

Item	Criterion	Value / finding	Conclusion
Temperature	> 37.8°C counts as fever	38.4°C	Abnormal · supports infection
Pharyngeal findings	Redness and swelling + exudate leans bacterial	Bilateral redness and swelling · white exudate	Abnormal · supports bacterial
Cervical lymph nodes	Tenderness and swelling count 1 point	Bilateral enlargement and tenderness	Abnormal · Centor +1
Cough	Cough present leans viral	None	Negative · Centor +1
RADT	Positive confirms GAS	Positive	GAS confirmed
WBC	> 10,000 supports bacterial	12,600 /μL	Abnormal · supports bacterial

Then the DiagnosisType funnel converges: layer one excludes red flags (no dyspnoea, no difficulty opening the mouth); layer two separates viral from bacterial (Centor 3 points); layer three confirms Group A Streptococcus by a positive RADT. Confirmed diagnosis: acute bacterial tonsillitis (GAS), high confidence.

5.4 Prescription Slip and Review Report

The prescription is produced in S5 and reviewed in S6. The prescription slip format is as follows (illustrative):

Prescription slip output (illustrative · demonstration only)

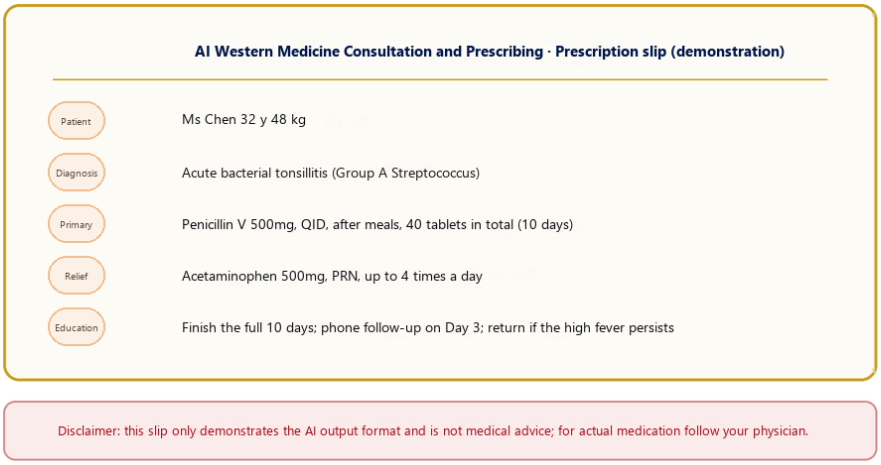


Figure 5-3 Prescription slip output format (demonstration only)

Table 5-4 Prescription slip (full version · demonstration only)

Component	Drug · content	Dose · route · frequency	Days · total quantity	Role marker
Primary drug	Penicillin V	500 mg · PO · QID	10 days · 40 tablets	Not freely replaceable
Symptom relief	Acetaminophen	500 mg · PO · PRN (≤ QID)	As needed · ≤ 40 tablets	Adjunct
Local treatment	Warm salt-water gargle	Every 4–6 hours	10 days	Adjunct
Patient education	Finish the course · return on red flags	—	10 days	Level-three component
Alternative plan	Azithromycin (if allergic)	500 mg · PO · QD	5 days · 5 tablets	Fallback plan

Table 5-5 Review report

Check item	What is checked	Result	Reason
Contraindication	Penicillin allergy · pregnancy and lactation · renal function	Pass	Allergy history negative · not pregnant or lactating · renal function normal
Interaction	Conflict with current medication	Pass	No long-term medication; acetaminophen does not conflict with the primary drug
Dose	Adjusted for weight and age	Pass	32 years old, 56 kg, adult standard dose
Release	All three questions pass	Release	May dispense

Disclaimer (3): the prescription in Table 5-4 and every drug dose in this book demonstrate output format only and do not constitute medical advice; for actual medication follow medical orders and the local drug regulations and latest clinical guidelines.

5.5 Follow-Up Dose Adjustment and Common Errors

The FollowType is produced in S7. In this case the Day 3 telephone follow-up showed fever resolved and pain reduced — a pass; the Day 10 visit showed the 10-day course completed and symptoms relieved, so the course is closed. Had the fever not resolved at Day 3, the pathway would be to push back to S4 and re-run the differential (complication? resistance? wrong diagnosis?) rather than switching straight to a stronger drug — this is exactly the value of the deductive skeleton: going back has a named stop.

Table 5-6 Common output errors in the case, compared

Erroneous output	Where it goes wrong	The correct approach
Asking 20 questions at once	Heavy patient burden, poor answer quality	S1 limits to 1–3 questions per round
RADT result read first, Centor score filled in afterwards	Data before criteria, reasoning fails	S3 writes the criteria before the values
The prescription contains only the primary drug	BOM missing parts, patient education missing	S5 brings out all four components
Day 3 without fever resolution → escalate the antibiotic directly	Skipping re-differentiation	S7 pathway pushes back to S4
A new table for the follow-up visit	The container breaks, identity drifts	Next page in the same container, id(C) unchanged

Table 5-7 S7 output excerpt: two education versions + red-flag list

Version	Content
Patient version (plain language)	Take the antibiotic four times a day for the full 10 days, and finish it even if you feel better; an unfinished course lets the bacteria come back. For the sore throat you may take a painkiller and gargle with warm salt water.
Physician version (brief)	Penicillin V 500 mg QID × 10 d, complete the course; acetaminophen PRN; saline gargle q4–6h.
Red-flag list (any one means return to clinic)	High fever above 39.5°C that will not settle, worsening difficulty swallowing, difficulty breathing or opening the mouth, a growing neck mass, or no improvement at all after 3 days.

Table 5-8 Follow-up record and dose-adjustment decision (time slice t → t+1)

Time slice	Findings	Criterion comparison	Decision
Day 3 telephone follow-up (t+1)	Fever gone, pain down from 8 to 3	Pass standard: fever gone · pain reduced	Keep the prescription, no adjustment
(assumed) no fever resolution at Day 3	Still 38.8°C, swallowing more painful	Fail: the pathway pushes back to S4	Re-run the differential (complication? resistance?) → complication means referral to ENT

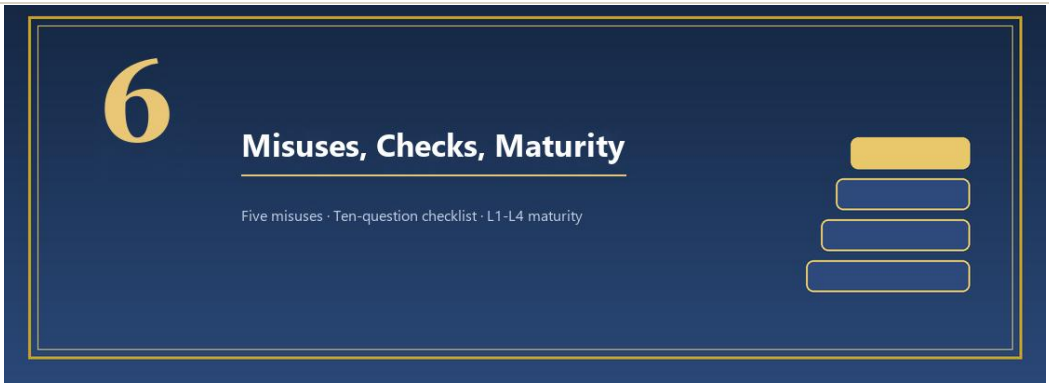
Day 10 case closed (t+2)	10-day course completed, symptoms relieved	Pass standard: medication taken throughout · symptoms relieved	Case closed and entered into the chain; tagged “first GAS episode” for future retrieval

Table 5-9 Rules for using the prompts

Rule	Explanation
Variables in place before starting	Run with the variables in 〔 〕 (history, data, baseline information) unfilled and the output will inevitably contain guessed values
Accept stage by stage, then relay	S1 output does not go into S2 before passing format acceptance; the iron law of garbage in, garbage out is doubly true inside a pipeline
Keep the original output	Every re-run is saved as a new version; the Type evidence chain must be able to replay “what was seen at the time”
Prompts may be tuned, the skeleton may not be abandoned	Fields and criteria may be adjusted by specialty, but the order — criteria before data, skeleton before prescription — does not move

Chapter focus

Chapter focus: (1) the S1–S7 seven-stage relay, each stage with clear, auditable inputs and outputs; (2) prompts with built-in format constraints and honesty clauses; (3) all case output follows criteria before data; (4) the prescription slip and the review report come out together; (5) follow-up dose adjustment needs criteria and a pathway, and a time slice appends only the differences.



Chapter 6 Misuses, Checks, Maturity

The last chapter delivers three governance tools: the misuse comparison table tells you where it will crash, the ten questions give you the acceptance Q&A, and the maturity model tells you which level you are on and what to add for the next.

6.1 Five Misuses

All five misuses share one feature: skipping the gate and adding the reason afterwards. The cure is not “being more careful” but writing the gate into the process — no pass, no output.

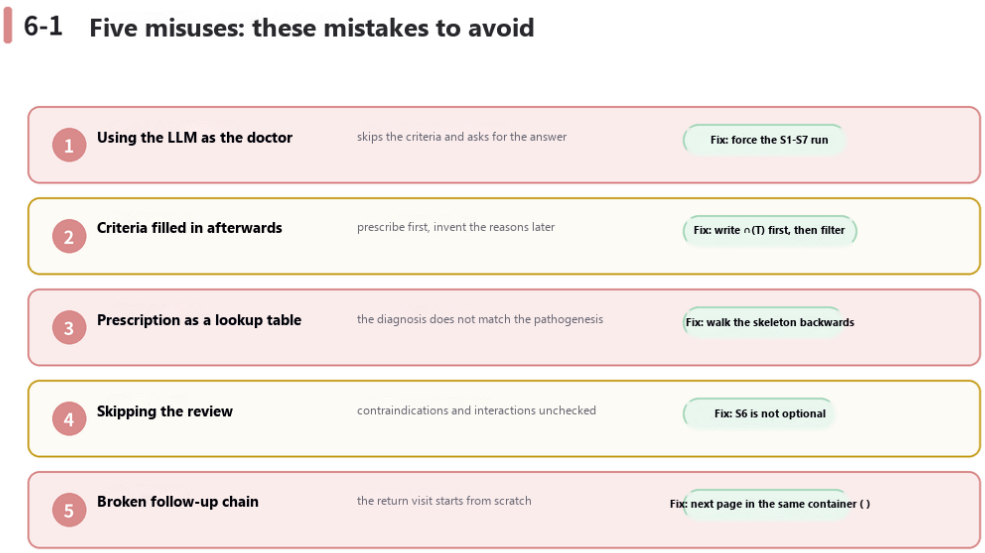


Figure 6-1 Five misuses and their remedies

Table 6-1 Misuse comparison table

Misuse	Typical presentation	Consequence	Remedy
Treating the LLM as the physician	Skipping S1–S7 and asking for the answer	No criteria, no evidence	Force the seven-stage pipeline through
Criteria patched afterwards	Prescribe first, fit the reason later	Reasoning disguised as reasoning	∩(T) first, order check
Prescription reduced to table lookup	The diagnosis does not match the pathogenesis	The same drug under a different disease name	Reverse walk of the deductive skeleton
Skipping the review	Contraindications and interactions unchecked	Medication safety event	S6 indispensable, three questions three states
The follow-up chain breaks	The follow-up visit starts from scratch	Fragmented history	Next page in the container · id(C) anchoring

6.2 A Ten-Question Checklist

Before delivering or accepting any AI Western-medicine consultation and prescribing output, interrogate it with these ten questions one by one; whatever cannot be answered is not yet qualified.

Table 6-2 The ten-question checklist

#	Question	What a passing answer looks like
1	What is the situation label?	Emergency / outpatient / chronic clearly stated and consistent with the complaint
2	Were the criteria written first?	The criterion timestamp precedes the data entry
3	Is the differential list broad enough?	At least 3 candidates, including red-flag exclusion
4	On what grounds does each funnel layer filter?	Every layer has criteria and the reason the excluded ones were excluded
5	Is the prescription BOM complete?	All four present: primary drug · relief · local · education
6	Does the deductive skeleton reach the bottom?	Cause → pathogenesis → evidence → principle → prescription, link by link
7	Are all three review questions answered?	Each of the three gives pass / caution / block plus a reason
8	Are level-three components marked?	Days · primary drug marked on record
9	Is the follow-up pathway written?	Every node has a pass standard and a failing pathway
10	Is the evidence chain complete?	Four auditable stages: input · intermediate · output · replay

6.3 Maturity L1–L4

Maturity has four levels: L1 answers, L2 screens, L3 follows up, L4 evolves. Upgrading is evidenced by the Type evidence chain; “it feels ready by now” is not accepted.

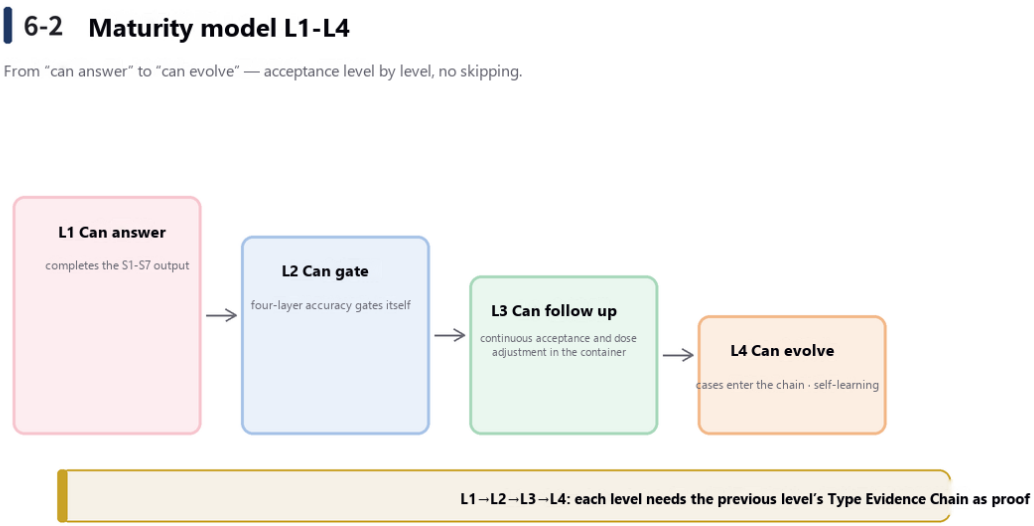


Figure 6-2 The maturity ladder: every level needs evidence

Table 6-3 Maturity acceptance table

Level	Capability	Acceptance condition	Common sticking point
L1 Answers	Can run the seven stages and produce the six Types	Six Types format-compliant, fields complete	Unstable output format
L2 Screens	Four-gate accuracy blocks automatically	The five misuses intercepted automatically	The review gets skipped
L3 Follows Up	Continuous acceptance and dose adjustment inside the container	All follow-up nodes executed	The follow-up chain breaks
L4 Evolves	Cases enter the chain · self-learning	Dose-adjustment reasons replayable, therapy refactoring grounded	Evolution unrecorded

Table 6-4 Rollout blueprint: a suggested three-month rhythm

Phase	Cycle	What to do	Completion criterion
Month 1 · get it running	W1–W4	Run the whole flow with a single disease (e.g. bacterial tonsillitis) along S1–S7	20 format-compliant outputs accumulated
Month 2 · put the gate in	W5–W8	Switch on four-gate accuracy acceptance; spot-check three outputs a day against the skeleton	Interception rate of the five misuses ≥ 90%
Month 3 · add follow-up	W9–W12	Switch on FollowType nodes and the dose-adjustment pathway; cases start entering the chain	Follow-up execution rate ≥ 80%, chain-break rate = 0
Afterwards · evolve	Ongoing	Monthly review of dose-adjustment reasons, refactor the therapy template	Enter the L4 acceptance loop

Review and Close

R-1 Book review · one diagram

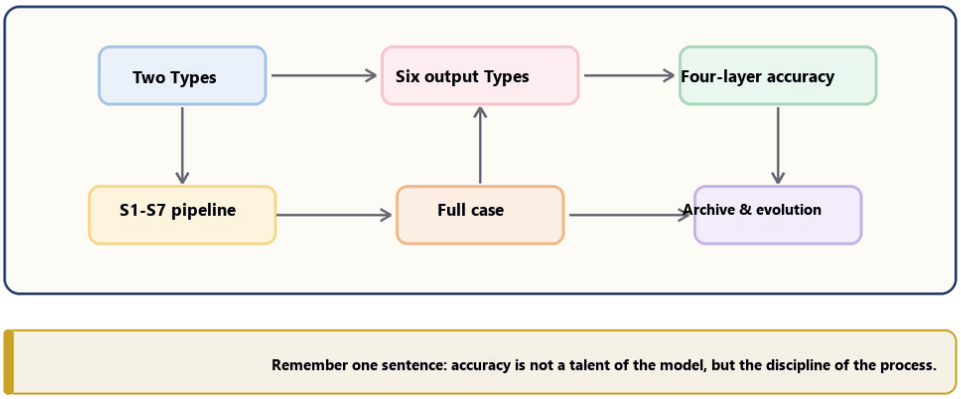


Figure R-1 The whole book in one diagram

Three takeaways

R-1 If you remember only one sentence: accuracy is not a talent of the model but the discipline of the process.

R-2 If you use only one table: the two-language comparison in Table D-1 — the Space Type builds structure, the Time Type issues moves.

R-3 If you run the process only once: Chapter 4 (six steps) plus S1–S7 from Chapter 5 (prompts and case), and work one complete output through by the book.

Table R-1 Master toolbox of the book (deliverables by chapter)

Chapter	Deliverable	What you can use straight away
Chapter 1 (understanding the system)	Positioning and where the two Types land	Type + Number division table, three-bracket landing table, four-gate table
Chapter 2 (output Types)	Specifications of the six output Types	Six-Type master specification, interview field card, prescription BOM specification
Chapter 3 (accuracy guarantee)	Four gates of accuracy	∩(T) criteria table, deductive skeleton table, prescription review checklist
Chapter 4 (six steps)	Operating pathway	Six-step operation card, three-layer judgement table, misuse warning points
Chapter 5 (prompts and case)	The full S1–S7 text + the case	Seven prompts, prescription slip example, follow-up dose-adjustment decision table
Chapter 6 (misuse checks)	Governance tools	Misuse comparison table, ten-question checklist, maturity acceptance table

A final reminder: every case, drug and dose in this book demonstrates output format and does not constitute medical advice; AI is a screening tool, not the prescribing subject — the prescribing right, and the responsibility, rest with a qualified physician.

Appendix A Quick Reference: Terms, Moves, Notation

A.1 Terminology Quick Reference

Table A-1 Quick-reference glossary

Term	English	In one sentence
Type	Type (not Form)	The structural unit that answers “what it is”
InterviewType	InterviewType	The structured output of history and complaint
ExamType	ExamType	The output of physical examination findings
LabType	LabType	The output of lab data interpretation
DiagnosisType	DiagnosisType	The output of the differential funnel and the confirmed diagnosis
Prescription	PrescriptionType	The output of the prescription BOM and the course
FollowType	FollowType	Nodes and pathways inside the course container
Intersection criterion	∩(T)	Data ∩ criteria; what fails is stopped
Identity anchor	id(C)	The guarantee that identity does not change inside the container
Three levels of remove	remove L1/L2/L3	Three-level grading of the consequences of removing components
Deductive skeleton	Deductive Skeleton	Cause → pathogenesis → evidence → principle → prescription
Type evidence chain	Type Evidence Chain	Traces at four stages: input · intermediate · output · replay

A.2 The Time Type Move Tray × Where It Lands in Consultation and Prescribing

Table A-2 Move tray comparison table

Move	Where consultation and prescribing land
Type-time function search	Retrieve similar cases from the history library
Design search and extraction	Extract prescribing rules from clinical guidelines
Design patterns	Prescription pattern library (first-line drug templates)
Logic generation and application	Rule generation for the differential funnel
Function generation and application	Follow-up function (nodes · assessment · pathway)
Continuous learning	New cases enter the chain, guidelines update
Refactoring and recombination	Therapy refactoring (second-line drug combination)
Problem solving	Breaking down a difficult case
Analysis	Symptom and lab interpretation
Design (creation and innovation)	Individualised course design
Reasoning (causal, reverse, etc.)	From symptoms to cause, from laboratory back to the source of infection
Thinking methods	Forward · reverse · vertical · first principles
R&D and exploration	Exploring new therapies and new combinations
Self-evolution	Dose-adjustment reasons replayed, the system improves itself

A.3 Notation Quick Reference

Table A-3 Notation table

Notation	Reading	Purpose
[ ]	Property Set · Abstract Set	X axis: symptom and lab-value fields
{ }	Composition Set · BOM	Y axis: prescription component composition
( )	Container Set	Z axis: body · course · identity
∩(T)	Intersection criterion	The filter gate between data and criteria
id(C)	Container identity	Identity unchanged within the course
remove	Removal grading	Grading of consequences after a component is removed
t / t+1	Time slice	Appending only the differences between first and follow-up visit

Accuracy comes from visible reasoning

Accuracy Comes From Visible Reasoning

Two-Type modelling

Space Type · three brackets × Time Type · move tray

Six output Types

Interview · Exam · Lab · Diagnosis · Prescription · Follow-up

Four-layer accuracy

Criterion funnel · Deductive skeleton · Prescription review · Evidence chain

S1-S7 pipeline

Full prompt text + end-to-end case output

